

Secure and Verifiable Optional-Dimension Data Aggregation for Efficient WBAN Deployment

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Abstract—Wireless Body Area Networks (WBANs) enable continuous monitoring of vital signs via wearable or implantable biosensors. To reduce cloud-side computation, terminal nodes locally process multidimensional physiological data and forward aggregated results to an intermediate aggregator. However, existing approaches face three key limitations: (1) the lack of terminal verification exposes WBANs to data authenticity risks, including tampering and lazy reporting; (2) privacy-preserving mechanisms such as secret sharing and homomorphic encryption incur high computational overhead, limiting efficiency and scalability; and (3) users are required to share all data dimensions at once, preventing personalized and flexible data sharing. To address these challenges, we propose a secure and verifiable optional-dimension data aggregation scheme SVODA for scalable WBAN deployment. The SVODA scheme integrates differential privacy to enhance protection without extra computational burden, and its effectiveness is validated through noise-error experiments on real-world datasets. It further incorporates a validation step at the aggregator to detect tampered data and identify non-responsive terminals. Moreover, SVODA allows users to selectively share specific data dimensions, enabling aggregation based on user consent rather than indiscriminate aggregation. Finally, we present a comprehensive security analysis and performance evaluation. The results demonstrate that our SVODA scheme is secure, while also achieving superior computational efficiency and storage performance compared to existing methods.

Index Terms—Wireless body area networks, data aggregation, differential privacy, verifiable data integrity.

I. INTRODUCTION

WIRELESS Body Area Networks (WBANs) [1] have become a vital enabler of next-generation healthcare by supporting continuous monitoring of physiological signals through wearable or implantable biosensors. These signals include heart rate, blood pressure, blood oxygen saturation, physical activity, sleep patterns, and fat burning data. The collected data can also be integrated into Electronic Health Records (EHRs) [2], enhancing communication between patients and healthcare providers, while aggregated WBAN data also supports clinical research and public health studies by enabling the identification of health trends, the development of new therapies, and disease prediction. For example, public health agencies can analyze the average body temperature of communities to estimate the incidence and spread of infectious diseases. Pharmaceutical research centers may evaluate drug efficacy by examining changes in physiological indicators over time following medication administration. In addition, hospitals can periodically assess aggregated physiological metrics of specific populations to adjust clinical strategies and optimize healthcare resource allocation. These capabilities make WBANs particularly important for chronic disease management, personal health tracking, and clinical decision support.

Cloud servers, with their strong computing and storage capabilities, can efficiently handle large-scale concurrent tasks such as homomorphic encryption, making them well-suited for storing and processing electronic health records (EHRs). However, this raises critical privacy concerns, as sensitive healthcare data may be exposed to the unauthorized access or inference. To mitigate these risks, various cryptographic schemes have been proposed [3], [4], [5]. One line of research employs Secret Sharing (SS) primitive [6], which divides sensitive information into multiple shares such that only a sufficient subset can reconstruct the original secret [7]. While this protects the privacy of physiological signals, it requires customized secure computation protocols for EHR analysis, and the intensive interaction among multiple parties results in substantial overhead. Another line of work leverages Homomorphic Encryption (HE) (e.g., Paillier) [8], enabling secure computation directly on ciphertexts [3], [4]. However, the reliance on costly exponentiation operations leads to heavy computational overhead. To strike

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a balance, hybrid protocols have been introduced by combining secret sharing with homomorphic encryption [9]. Although these schemes effectively safeguard data privacy, the use of SS and/or HE inevitably introduces computational overhead several orders of magnitude higher than plaintext computation [5], making them inefficient and difficult to scale in real-world WBAN deployments. This leads to the **first challenge**: *how to ensure privacy protection without introducing prohibitive overhead*.

Another limitation of existing schemes is the lack of mechanisms to guarantee data authenticity and integrity. Most research assumes that terminal nodes behave honestly [9], [10], yet in practice, devices may be compromised or behave lazily, leading to tampered reports or missing data contributions [7]. Without effective verification, such faulty or malicious behavior can distort aggregation results and undermine trust in the system. This highlights the **second challenge**: *how to design an aggregation process that can validate data integrity and identify non-responsive or malicious terminals*.

To alleviate the burden on cloud servers, the concept of fog computing has been introduced, enabling intermediate aggregating and processing at nodes closer to the data sources, thereby reducing latency and supporting real-time analysis [11], [12]. However, most existing studies focus on one-dimensional data aggregation, which offers simplicity and low computational cost [13], but fails to capture the comprehensive state of human health. To overcome this limitation, multidimensional aggregation schemes [3], [4] enable terminal nodes to locally process multiple physiological signals before forwarding them to the aggregator, thereby reducing bandwidth consumption. Nevertheless, current multidimensional schemes typically require users to share all data dimensions once they agree to participate, overlooking their personalized sharing preferences [11], [14]. For example, an obese patient may be willing to share blood pressure readings but reluctant to disclose weight data. This raises the **third challenge**: *how to reduce bandwidth consumption while supporting user-centric, flexible multidimensional data sharing*.

To address these challenges, this paper proposes a Secure and Verifiable Optional-Dimension Data Aggregation (SVODA) scheme for WBANs. SVODA incorporates differential privacy at the terminal nodes to preserve privacy without incurring heavy cryptographic overhead. It further introduces a lightweight validation mechanism at the aggregator, which verifies sub-shares during secret reconstruction to detect tampered data or lazy terminals, thus ensuring data authenticity. Moreover, by supporting optional-dimension aggregation, SVODA allows users to selectively share specific data dimensions, thereby accommodating personalized sharing preferences. In summary, SVODA simultaneously achieves privacy protection, data integrity, and communication efficiency, offering a scalable solution for secure WBAN deployment. Our contributions are fourfold and can be described as follows.

- We incorporate differential privacy at the data terminal nodes to prevent adversaries from inferring sensitive physiological information, significantly reducing the computational burden compared with traditional encryption-based schemes. Meanwhile, theoretical and experimental proofs

of the noise error are conducted to verify practicality and accuracy.

- We propose a lightweight secret-sharing-based validation mechanism that supports optional-dimensional data aggregation. By embedding verification values into secret shares, the proposed approach enables the aggregator to detect tampered data or identify non-responsive (lazy) terminals, while allowing users to selectively disclose specific data dimensions according to personalized sharing preferences. This design ensures aggregation integrity and reliability and better fits flexible real-world WBAN applications.
- We conduct a detailed theoretical analysis and extensive experimental evaluation of the proposed SVODA scheme. The results demonstrate that SVODA achieves privacy protection and robustness while significantly outperforming existing approaches in both computational efficiency and communication performance.

The remainder of this paper is structured as follows. Section II begins with a review of the necessary preliminaries, followed by Section III, which introduces the problem formulation. We then describe the proposed SVODA scheme in Section IV and provide its formal security proof in Section V. Section VI moves on to the performance evaluation and comparison results. Section VII discusses related work. Finally, Section VIII concludes the paper.

II. PRELIMINARIES

A. Multisecret Sharing Scheme

Multisecret sharing scheme [15] allows a secret holder to simultaneously share multiple secrets (e.g., k secrets) among n participants. This scheme is based on a threshold parameter t , such that any subset of at least t participants can jointly reconstruct all the k secrets, while any subset of fewer than t participants gains no information about the secrets. The parameters satisfy the condition $1 \leq k < t \leq n$. The following construction is an extension of Shamir's (t, n) threshold secret sharing scheme [6], adapted to support multiple secrets. The detailed steps are described as follows.

- Suppose there is a secret holder who divides the secret s into k parts, denoted as $\{s_0, \dots, s_{k-1}\} \in GF(p)$ and $k \leq p$, with n participants. Then, the holder sets t as the threshold for recovering this secret. First, the holder determines a finite field $GF(p)$, and then generates a polynomial as follows:

$$y = f(x) = \left(\sum_{i=0}^{k-1} s_i \cdot x^i + \sum_{i=0}^{t-1-k} r_i \cdot x^{i+k} \right) \bmod p.$$

The coefficients before x^0 to x^{k-1} in this polynomial are secrets s_0 to s_{k-1} , and the coefficients before x^k to x^{t-1} are random numbers r_i selected by the holder.

- The holder then inputs the x_i of different participants into the polynomial to generate y_i , and sends this coordinate (x_i, y_i) to the participants. As long as there are t such coordinate points, a $(t-1)$ -order polynomial can be uniquely determined.

- Secrets can be recovered from any t different participants through Lagrange interpolation polynomial. Using the following formula:

$$f(x) = \sum_{i=0}^{t-1} \left(y_i \cdot \prod_{j=0, j \neq i}^{t-1} \frac{x - x_j}{x_i - x_j} \right) \bmod p.$$

Homomorphism: The Lagrange interpolation polynomial is homomorphic and can be described in detail as follows: Let $f(x)$ and $g(x)$ be n -degree polynomials. The sets of points $P_1(x_i, y_i)$ and $P_2(x_i, y'_i)$ determine the polynomials $f(x)$ and $g(x)$ through the Lagrange interpolation method. Consequently, the set of points $P_{12} = \{x_i, (y_i + y'_i) | 0 \leq i \leq n\}$ can determine the polynomial that is the sum of $f(x)$ and $g(x)$.

B. Differential Privacy

ϵ -Differential Privacy [16]: For any two datasets D_u and D_v that differ by a single element, and for all subsets R in the range of the algorithm $\mathcal{A}$, the probability that the output of $\mathcal{A}$ on dataset D_u falls within R is at most e^ϵ times the probability that the output of $\mathcal{A}$ on dataset D_v falls within R . Thus, the randomized algorithm $\mathcal{A}$ is said to achieve ϵ -Differential Privacy:

$$\Pr[\mathcal{A}(D_u) \in R] \leq e^\epsilon \cdot \Pr[\mathcal{A}(D_v) \in R].$$

$\mathcal{L}_1$ -Sensitivity: The $\mathcal{L}_1$ -sensitivity of a query function Q is defined as the maximum $\mathcal{L}_1$ distance between the outputs of Q on any two neighboring datasets D_u and D_v . It quantifies the largest possible change in the output of Q resulting from the modification of a single element in the dataset. Formally,

$$\Delta Q = \max_{D_u, D_v} \|Q(D_u) - Q(D_v)\|_1,$$

where D_u and D_v differ by at most one element.

Laplace Mechanism: The Laplace mechanism ensures differential privacy for query functions by adding independent and identically distributed Laplace noise to the exact query outputs. Formally, let $Lap(\lambda)$ represents a random variable with a Laplace distribution with probability density function $\Pr(x, \lambda) = \frac{1}{2\lambda} \cdot e^{-\frac{|x|}{\lambda}}$. In this mechanism, noise is sampled from $Lap(0, \lambda)$, which has mean 0 and scale parameter $\lambda = \frac{\Delta Q}{\epsilon}$. Intuitively, queries with higher sensitivity require larger noise to ensure the same privacy guarantee.

Infinite Divisibility of Laplace Distribution: In distributed differential privacy, each user perturbs their own data to protect individual privacy, rather than relying solely on a centralized noise generator. This is enabled by the infinite divisibility of the Laplace distribution, which allows a Laplace random variable to be expressed as the sum of independent components. Specifically, $L \sim Lap(\theta, \lambda)$ can be represented as the sum of $2n$ gamma random variables [17]:

$$L = \frac{\theta}{n} + \sum_{i=1}^n (G_{1,i}(n, \lambda) - G_{2,i}(n, \lambda)),$$

where θ denotes the location parameter of the Laplace distribution, $G_{1,i}(n, \lambda)$ and $G_{2,i}(n, \lambda)$ are independent gamma random

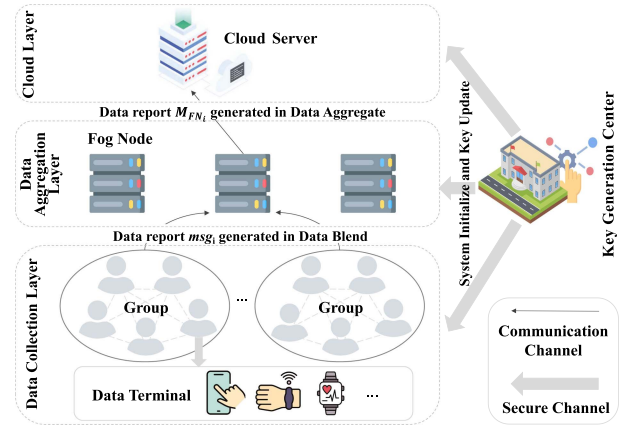


Fig. 1. System model of the proposed scheme.

variables with probability density:

$$\Pr(G_{1,i} = x) = \Pr(G_{2,i} = x) = \frac{\left(\frac{1}{\lambda}\right)^{\frac{1}{n}}}{\Gamma\left(\frac{1}{n}\right)} x^{\frac{1}{n}-1} e^{-x/\lambda},$$

with $\Gamma(\cdot)$ denoting the gamma function. Through this decomposition, users locally generate partial noise, and their aggregation yields a Laplace-distributed result, ensuring ϵ -differential privacy in a decentralized setting.

III. PROBLEM FORMULATION

This section comprises four parts: the system model, which specifies all entities and their interactions; the threat model, which characterizes the adversary's capabilities; the design goals, which state the security and efficiency goals; and, finally, the formal syntax of the proposed SVODA scheme.

A. System Model

The system architecture is a typical three-layer structure architecture, mainly including the *Data Collection Layer*, *Data Aggregation Layer*, *Cloud Server Layer* and *Key Generation Center*, as shown in Fig. 1. The specific details are described as follows:

1) **Data Collection Layer:** This layer consists of multiple data terminals $T_i^{u,g}$, where $1 \leq i \leq n$ denotes the terminal index, $1 \leq u \leq w$ denotes the group index, and g specifies fog node. Each terminal corresponds to a user-worn or ambient sensing device, such as a wristband or medical monitor, that collects vital sign data from user. In SVODA, terminals $T_i^{u,g}$ autonomously form groups based on users' social relationships, such as family, friends, or colleagues. With the user's consent, these terminals periodically collect and process personal health data, and subsequently transmit the aggregated reports to the corresponding fog node FN_g .

2) **Data Aggregation Layer:** This layer consists of several fog nodes FN_g ($1 \leq g \leq z$), which, due to their proximity to data sources, provide real-time processing and low latency compared with architectures lacking a fog layer. In SVODA, each FN_g receives data from terminals (i.e., users' devices),

reconstructs and verifies data for each group. Then, FN_g generates both aggregated results and dimension-specific selection data across multiple groups.

3) *Cloud Server Layer*: This layer consists of a cloud server (CS), which possesses extensive storage and computational resources across networks and geographic regions. In SVODA, CS receives aggregated data from fog nodes FN_g , decrypts them, and obtains the final multidimensional selection and aggregation summaries for users. Based on these results, CS can further compute statistical indicators such as the mean value of each medical data dimension. In this paper, we assume that CS is an honest but curious entity that does not have cloud monitoring capabilities.

4) *Key Generation Center*: Key Generation Center (KGC) is responsible for system initialization and key update. It generates the necessary cryptographic keys and verification parameter for SVODA and securely distributes them to the relevant entities via a secure channel. In this paper, only KGC distributes keys through secure channels for transmission, as indicated by the arrows labeled in Fig. 1.

B. Threat Model

In SVODA, We follow the trend of considering KGC as trustworthy [4], [18], [19]. KGC strictly follows the established protocol, providing registration services and distributing keys. The cloud server and fog nodes are regarded as semi-trusted entities that may attempt to access users' sensitive information and could potentially collude with each other. Data terminals may behave lazily and fail to submit data to the aggregation nodes. Additionally, the possible attacks under this threat model include:

- 1) *Differential Attack*: An adversary leverages users' background knowledge and analyzes the differences between grouped outputs from consecutive queries to infer the private information of a specific user.
- 2) *Replay Attack*: An adversary intercepts valid communication messages and retransmits them to the target in order to impersonate legitimate user behavior.
- 3) *Man-in-the-Middle Attack*: An adversary positions itself between communicating parties to eavesdrop on or tamper with transmitted data, making both parties believe they are communicating directly with each other.
- 4) *Data Tampering*: An adversary intercepts users' private information and maliciously alters it before uploading to the aggregator, leading to inaccurate final aggregation results.

C. Design Goals

This paper is committed to designing a privacy preserving multi-dimensional data aggregation protocol, and our protocol can achieve the following design goals:

- 1) *Correctness*: The scheme must ensure that aggregation results are accurate and verifiable. Data from terminals should be correctly incorporated, fog nodes must validate its integrity, and the cloud server should obtain the true final results after decryption, ensuring reliable statistical analysis.

2) *Data Utility*: To ensure practical utility, the scheme should maintain data availability after applying differential privacy, striking a balance between privacy protection and utility. Specifically, the error introduced by noise should remain relatively small and stable, thereby preventing degradation of data usability for medical analysis and decision-making and ensuring that the aggregated outputs remain meaningful for healthcare applications.

3) *Privacy Protection*: Biometric and health-related data are highly sensitive and must remain private. The SVODA scheme should ensure confidentiality during transmission and storage, enforcing strict access controls, and resisting common attacks such as differential and replay attacks.

4) *Data Integrity*: In order to prevent malicious or lazy nodes from tampering with data and protect data integrity, aggregation fog nodes need to perform verification operations before receiving data to ensure that the data has not been tampered with and that there are no lazy nodes in the system that do not send data information.

5) *Fault-Tolerant*: Real-world systems are often subject to hardware and software failures, and the likelihood of terminal failure (e.g., device malfunction or network disruption) is uncertain. Therefore, the proposed scheme should incorporate fault tolerance mechanisms to ensure system reliability even when individual components fail.

6) *Resist Collusion Attack*: A potential risk in any distributed system is the possibility of collusion between fog nodes and cloud servers, which could compromise user privacy by facilitating unauthorized access to sensitive information. In such scenarios, our scheme should be to prevent collusion between fog nodes and servers.

D. Syntax

We formally define the proposed SVODA scheme, which is composed of the following five protocols: *System Initialize*, *Key Update*, *Data Blend*, *Data Aggregate*, and *Result Report*. Table I summarizes the main notations used in the SVODA scheme. The high-level syntax descriptions of each phase are provided below.

1) *System Initialize*: This protocol consists of three algorithms: Parameter Initialization (Paralnit), Entity Initialization (Entitylnit) and Group Initialization (Grouplnit).

- $\text{Paralnit}(1^{\lambda_{sec}}) \rightarrow par$: The parameter initialization algorithm is performed by KGC , inputs a security parameter λ_{sec} , then outputs the system public parameter $par = \{\mathbb{N}, H, p\}$, where $\mathbb{N}$ is a large prime number, $H : \{0, 1\}^* \rightarrow \mathbb{Z}_{\mathbb{N}}^*$ is a secure hash function, p is a prime number which is bigger than $\mathbb{N}$.
- $\text{Entitylnit}(par) \rightarrow (x_i^{u,g}, y_g, sk_i^{u,g}, sk_g, k_i^{u,g}, k_g)$: The entity initialization algorithm is performed by KGC , inputs the system public parameter par , and then outputs the private key $sk_i^{u,g}$, sk_g , and $x_i^{u,g}, y_g \in \mathbb{Z}_{\mathbb{N}}^*$ as the identity information for the data terminal $T_i^{u,g}$ and fog node FN_g , and $k_i^{u,g}, k_g$ used for message authentication.
- $\text{Grouplnit}(G_u^g) \rightarrow (t_u^g, (a_{u,0}^g, a_{u,1}^g, \dots, a_{u,n}^g))$: The group initialization algorithm is performed by KGC , inputs the set of groups $G_u^g \sim (1 \leq u \leq w)$ organized by data terminals

TABLE I
DESCRIPTIONS OF NOTATIONS

Notations	Descriptions
f	The total number of data terminals
n	The number of members in a group
t_u^g	The threshold of the secret sharing scheme in the u -th group G_u^g which the g -th fog node belongs
m	The number of data dimensions
w	The number of groups corresponding to a FN_g
z	The number of fog nodes corresponding to a CS
$x_i^{u,g}$	The identity of data terminal $T_i^{u,g}$
y_g	The identity of fog node FN_g
$sk_i^{u,g}$	The key between $T_i^{u,g}$ and CS
$v_{i,j}^{u,g}$	The one-time key of j -th element in $T_i^{u,g}$
$a_{u,i}^g$	The parameters used for verification
ts	Timestamp
$d_{i,j}^{u,g}$	The j -th element of $T_i^{u,g}$
$s_{i,j}^{u,g}$	The selection for the j -th element of $T_i^{u,g}$
$md_{i,j}^{u,g}$	The j -th element of $T_i^{u,g}$ with added perturbation
$c_{d_{i,j}^{u,g}}$	Ciphertexts of DP-perturbed data $md_{i,j}^{u,g}$
$c_{s_{i,j}^{u,g}}$	Ciphertexts of selection data $s_{i,j}^{u,g}$
M_j	Aggregation plaintext of j -th element
S_j	Aggregation plaintext of j -th selection element

$T_i^{u,g}$, and then outputs group SS threshold t_u^g and verification parameter $(a_{u,0}^g, a_{u,1}^g, \dots, a_{u,n}^g)$.

2) *Key Update*: This protocol only contains a KeyUpdate algorithm. The specific process is as follows.

- $\text{KeyUpdate}(sk_i^{u,g}, x_i^{u,g}) \rightarrow (v_{i,j}^{u,g}, v_{i,m+j}^{u,g}, SK_j, SK_{m+j})$: The key update algorithm is performed by KGC , inputs the private key $sk_i^{u,g}$ and identity information $x_i^{u,g}$, outputs the private key $v_{i,j}^{u,g}$ and $v_{i,m+j}^{u,g}$ for $T_i^{u,g}$, SK_j and SK_{m+j} for each element $j \in [1, m]$.

3) *Data Blend*: This protocol consists of two following algorithms: DataProcessing and SecShareGen.

- $\text{DataProcess}(D_i^{u,g}, S_i^{u,g}) \rightarrow (md_{i,j}^{u,g}, c_{d_{i,j}^{u,g}}, c_{s_{i,j}^{u,g}})$: The data processing algorithm is performed by $T_i^{u,g}$, inputs the users' healthcare sensor dataset $D_i^{u,g}$ and the selection set $S_i^{u,g}$, then output the DP-perturbed data $md_{i,j}^{u,g}$ and briefly encrypted ciphertext $c_{d_{i,j}^{u,g}}$ and $c_{s_{i,j}^{u,g}}$.
- $\text{SecShareGen}(c_{d_{i,j}^{u,g}}, c_{s_{i,j}^{u,g}}, a_{u,i}^g) \rightarrow (msg_{i,l}^{u,g}, msg_{i,i}^{u,g}, tag_i^{u,g})$: The secret-share generation algorithm is performed by $T_i^{u,g}$, inputs the ciphertext $c_{d_{i,j}^{u,g}}$, $c_{s_{i,j}^{u,g}}$ and the verification value $a_{u,i}^g$, then outputs the data report $msg_{i,l}^{u,g}$ for other group members and $msg_{i,i}^{u,g}$ for FN_g , and $tag_i^{u,g}$ used for message authentication.

4) *Data Aggregate*: This protocol consists of two following algorithms: DataAggregate and FaultLocalization.

- $\text{DataAggregate}(I_{G_u}^g, y_g, a_{u,0}^g, msg_{i,i}^{u,g}) \rightarrow (V_{G_u}, V_{FN_g}, M_{FN_g}, tag_g)$: The data aggregation algorithm is performed by FN_g , inputs the set of groups $I_{G_u}^g$, the value $a_{u,0}^g$ and the data report $msg_{i,i}^{u,g}$, outputs the aggregated data V_{G_u} for each group and V_{FN_g} for multiple groups in FN_g and the data report M_{FN_g} . If the verification does not hold, FN_g will send the set of all shares $\mathbb{F}_u^g$ to KGC .

- $\text{FaultLocalization}(\mathbb{F}_u^g, \mathbb{G}, t_u^g) \rightarrow (\mathbb{V}_u^g, \mathbb{M}_u^g)$: The FaultLocalization algorithm is performed by KGC , inputs the set of groups $\mathbb{G}$, all shares in the group $\mathbb{F}_u^g$ and the threshold t_u^g , outputs valid data terminal set $\mathbb{V}_u^g$ and invalid data terminal set $\mathbb{M}_u^g$.

5) *Result Report*: This protocol only contains a ResReport algorithm. The specific process is as follows.

- $\text{ResReport}(M_{FN_g}, SK_j, SK_{m+j}) \rightarrow (M_j, S_j, Avg(M_j))$: This algorithm is performed by CS , inputs the data report M_{FN_g} , the private key SK_j and SK_{m+j} , outputs the decrypted aggregated data M_j , the selection data S_j and mean value $Avg(M_j)$ for each element $j \in [1, m]$.

IV. THE PROPOSED SCHEME

In this section, we present SVODA, a secure and verifiable optional-dimension data aggregation scheme tailored for intelligent healthcare systems. The scheme is organized into five phases: *System Initialize*, *Key Update*, *Data Blend*, *Data Aggregate*, and *Result Report*. Each phase plays a distinct role in ensuring privacy protection, data integrity, and efficient aggregation, with execution protocols detailed below.

A. System Initialize

This protocol consists of three following algorithms: Parameter Initialization (Paralnit), Entity Initialization (Entitylnit) and Group Initialization (Grouplnit), as shown in **Protocol 1**.

1) Paralnit: At this algorithm, KGC fulfills the agreement, keep its promises, and generate necessary system parameters for other entities in the healthcare system.

2) Entitylnit: KGC provides registration services for data terminals and fog nodes. Specifically, it generates the identity information $x_i^{u,g}$ and y_g for the data terminal $T_i^{u,g}$ and fog node FN_g . In addition, a random number $sk_i^{u,g} \in \mathbb{Z}_N^*$ is picked and defined as the key of $T_i^{u,g}$. And KGC sets the key $k_i^{u,g}$ and k_g for message authentication.

3) Grouplnit: Multiple $T_i^{u,g}$ spontaneously form a group G_u^g ($1 \leq u \leq w$) based on geographic location information or mutual trust, such as social circles of relatives, friends, colleagues, classmates, etc., which will serve as the scope for secret sharing. $T_i^{u,g}$ in the group negotiates a threshold t_u^g for secret sharing and secret recovery. The aggregation process is independently executed in each round, and the number of participating users in the new round can be dynamically adjusted. The threshold for secret sharing will also be renegotiated. KGC will choose random numbers $\{a_{u,1}^g, a_{u,2}^g, \dots, a_{u,n}^g\}$, and compute $a_{u,0}^g$ (Line 3), then send $a_{u,0}^g$ to FN_g and send $\{a_{u,1}^g, a_{u,2}^g, \dots, a_{u,n}^g\}$ to the corresponding $T_i^{u,g}$ ($1 \leq i \leq n$) in the group, respectively.

B. Key Update

The key update protocol only contains a KeyUpdate algorithm, as shown in **Protocol 2**.

The KGC selects a random number $\alpha_i^{u,g} \in \mathbb{Z}_N^*$ to compute a one-time password $v_i^{u,g}$. Then KGC computes j -th element key

Protocol 1: System Initialize.

ParalNit(1_{sec}^λ) $\rightarrow$ par

- 1: Define $\mathbb{N}$ is a prime number larger than f and m , p is a prime number larger than $\mathbb{N}$, $H : \{0, 1\}^* \rightarrow \mathbb{Z}_\mathbb{N}^*$ is a secure hash function. Finally, the system parameters is defined as $par \leftarrow (p, H, \mathbb{N})$

EntityNit(par) $\rightarrow (x_i^{u,g}, y_g, sk_i^{u,g}, sk_g, k_i^{u,g}, k_g)$

- 1: KGC picks random numbers $x_i^{u,g}, y_g \in \mathbb{Z}_\mathbb{N}^*$ to define the identifiers information of data terminal $T_i^{u,g}$ and fog node FN_g
- 2: Selects a random number $sk_i^{u,g} \in \mathbb{Z}_\mathbb{N}^*$ as the private key of data terminal $T_i^{u,g}$
- 3: Sets $k_i^{u,g} \leftarrow H(sk_i^{u,g} || \text{"MAC"})$
- 4: Sends $k_i^{u,g}$ to data terminal $T_i^{u,g}$ and fog node FN_g
- 5: Selects a random number $sk_g \in \mathbb{Z}_\mathbb{N}^*$
- 6: Sets $k_g \leftarrow H(sk_g || \text{"MAC"})$
- 7: Sends k_g to fog node FN_g and CS

GroupNit(G_u^g) $\rightarrow (t_u^g, (a_{u,0}^g, a_{u,1}^g, \dots, a_{u,n}^g))$

- 1: All data terminals in G_u^g negotiate a threshold value $t_u^g \in [1, n]$
- 2: KGC chooses random numbers $\{a_{u,1}^g, a_{u,2}^g, \dots, a_{u,n}^g\} \in \mathbb{Z}_\mathbb{N}^*$ for data terminals in the group G_u^g
- 3: Computes verification value $a_{u,0}^g = (a_{u,1}^g + \dots + a_{u,n}^g) \bmod \mathbb{N}^2$
- 4: Sends $a_{u,0}^g$ to the fog nodes FN_g
- 5: Sends $\{a_{u,1}^g, a_{u,2}^g, \dots, a_{u,n}^g\}$ to the corresponding $T_i^{u,g}$ ($1 \leq i \leq n$) in the group G_u^g , respectively

Protocol 2: Key Update.

KeyUpdate($sk_i^{u,g}, x_i^{u,g}$) $\rightarrow (v_{i,j}^{u,g}, v_{i,m+j}^{u,g}, SK_j, SK_{m+j})$

- 1: KGC choose a random value $\alpha_i^{u,g} \in \mathbb{Z}_\mathbb{N}^*$
- 2: Compute one-time password $v_i^{u,g} \leftarrow H(\alpha_i^{u,g} || sk_i^{u,g} || x_i^{u,g})$
- 3: **for** $j = 1, 2, \dots, m$ **do**
- 4: Compute $v_{i,j}^{u,g} \leftarrow H(v_i^{u,g} || sk_i^{u,g} || x_i^{u,g} || j)$
- 5: Compute $v_{i,m+j}^{u,g} \leftarrow H(v_i^{u,g} || sk_i^{u,g} || x_i^{u,g} || m + j)$
- 6: **end for**
- 7: Compute $SK_j = \sum_{i=1}^f v_{i,j}^{u,g}$
- 8: Compute $SK_{m+j} = \sum_{i=1}^f v_{i,m+j}^{u,g}$
- 9: Send $v_{i,j}^{u,g}$ and $v_{i,m+j}^{u,g}$ to $T_i^{u,g}$
- 10: Send SK_j and SK_{m+j} to CS
- 11: **return**

$v_{i,j}^{u,g}$ and $(m+j)$ -th key $v_{i,m+j}^{u,g}$ for encrypting data and selection based on different dimensions(line 3-6). And KGC adds all keys to obtain j -dimensions key aggregation SK_j and $(m+j)$ -dimensions key aggregation SK_{m+j} used for decryption(line 7-8). Finally, KGC sent $v_{i,j}^{u,g}, v_{i,m+j}^{u,g}$ to $T_i^{u,g}$ and send SK_j, SK_{m+j} to CS through a secure channel.

Protocol 3: Data Blend.

DataProcessing($D_i^{u,g}, S_i^{u,g}$) $\rightarrow (md_{i,j}^{u,g}, c_{d_{i,j}}^{u,g}, c_{s_{i,j}}^{u,g})$

- 1: Extracts $d_{i,j}^{u,g}$ and $s_{i,j}^{u,g}$ from set $D_i^{u,g}$ and $S_i^{u,g}$
- 2: **for** $j = 1, 2, \dots, m$ **do**
- 3: Extracts partial noise from Gamma distribution $G_{i,j,1}(n, \lambda) - G_{i,j,2}(n, \lambda)$
- 4: Computes $md_{i,j}^{u,g} \leftarrow d_{i,j}^{u,g} + G_{i,j,1}(n, \lambda) - G_{i,j,2}(n, \lambda)$
- 5: Computes $c_{d_{i,j}}^{u,g} \leftarrow (s_{i,j}^{u,g} \cdot md_{i,j}^{u,g} + v_{i,j}^{u,g}) \bmod \mathbb{N}$
- 6: Computes $c_{s_{i,j}}^{u,g} \leftarrow (s_{i,j}^{u,g} + v_{i,m+j}^{u,g}) \bmod \mathbb{N}$
- 7: **end for**
- 8: **return** $md_{i,j}^{u,g}, c_{d_{i,j}}^{u,g}, c_{s_{i,j}}^{u,g}$

SecShareGen($c_{d_{i,j}}^{u,g}, c_{s_{i,j}}^{u,g}, a_{u,i}^g$) $\rightarrow (msg_{i,l}^{u,g}, msg_i^{u,g}, tag_i^{u,g})$

- 1: Record a current timestamp ts
- 2: **for** $i = 1, 2, \dots, n$ **do**
- 3: $T_i^{u,g}$ chooses random numbers $r_{i,0}^{u,g}, \dots, r_{i,k}^{u,g}$ ($0 \leq k \leq t_u^g - 2 - 2m$).
- 4: Computes $f_i^{u,g}(x)$ using (1).
- 5: **for** $l = 1, 2, \dots, i, \dots, n$ **do**
- 6: Computes $f_i^{u,g}(x_l^{u,g})$ based on $x_l^{u,g}$
- 7: Generates $msg_{i,l}^{u,g} \leftarrow x_i^{u,g} || ts || f_i^{u,g}(x_l^{u,g})$
- 8: Sends data report $msg_{i,l}^{u,g}$ to other data terminals $T_l^{u,g}$
- 9: $T_i^{u,g}$ verify the received data report $msg_{i,l}^{u,g}$
- 10: **end for**
- 11: Computes $f(x_i^{u,g}) = \sum_{l=1}^n f_l(x_i^{u,g}) \bmod p^2$
- 12: Computes $tag_i^{u,g} \leftarrow MAC_{k_i^{u,g}}(x_i^{u,g} || ts || f(x_i^{u,g}))$
- 13: Generates $msg_i^{u,g} \leftarrow x_i^{u,g} || ts || f(x_i^{u,g}) || tag_i^{u,g}$
- 14: Sends data report $msg_i^{u,g}$ to the corresponding fog node FN_g .
- 15: **end for**
- 16: **return**

C. Data Blend

This protocol consists of two following algorithms: DataProcessing and SecShareGen, as shown in **Protocol 3**.

1) DataProcessing: In this phase, $T_i^{u,g}$ periodically collects the user's healthcare data $d_{i,j}^{u,g}$ from sensor devices. The resulting m -dimensional data is represented as $D_i^{u,g} = (d_{i,1}^{u,g}, d_{i,2}^{u,g}, \dots, d_{i,m}^{u,g})$. Based on the user's sharing preferences, a selection vector is defined as $S_i^{u,g} = (s_{i,1}^{u,g}, s_{i,2}^{u,g}, \dots, s_{i,m}^{u,g})$, where $s_{i,j}^{u,g} = 1$ indicates that the user consents to share the data of that dimension, and $s_{i,j}^{u,g} = 0$ indicates that the user opts not to share the data due to privacy or other concerns. Finally, $T_i^{u,g}$ performs differential privacy processing generate $md_{i,j}^{u,g}$ (Line 4), then encrypts $md_{i,j}^{u,g}$ and $s_{i,j}^{u,g}$ to generate $c_{d_{i,j}}^{u,g}$ and $c_{s_{i,j}}^{u,g}$ (Line 5-6).

2) SecShareGen: In this phase, $T_i^{u,g}$ records the current timestamp ts . Specifically, $T_i^{u,g}$ collaboratively generate sub secrets required for secret sharing, which are used by FN_g to reconstruct the original secret. $T_i^{u,g}$ selects random numbers $r_{i,k}^{u,g} \sim (0 < k \leq t_u^g - 1 - 2m)$ to construct a polynomial

$f_i^{u,g}(x)$. As shown in (1)

$$f_i^{u,g}(x) = \left(r_{i,0}^{u,g} + \sum_{j=1}^m c_{d_{i,j}^{u,g}} x^j + \sum_{j=1}^m c_{s_{i,j}^{u,g}} x^{m+j} + \sum_{k=1}^{t_u^g-1-2m} r_{i,k}^{u,g} x^{2m+k} + a_{u,i}^g x^{t_u^g-1} \right) \bmod p \quad (1)$$

Among them, $(t_u^g - 1 - 2m)$ is the number of random number terms remaining in $(t_u^g - 1)$ -order polynomial after removing $2m$ ciphertext terms and one verification parameter term. Simultaneously, the user's multidimensional data and a verification parameter $a_{u,i}^g$ are embedded into the polynomial (Line 4), where $a_{u,i}^g$ is value to which data terminal $T_i^{u,g}$ belongs in group G_u^g . Next, $T_i^{u,g}$ computes $f_i^{u,g}(x_l)$ for the remaining data terminals $T_l^{u,g} (1 \leq l \leq n, l \neq i)$, packages the message into $msg_{i,l}^{u,g}$ and send it to respective terminals $T_l^{u,g}$. Upon receiving these messages, $T_i^{u,g}$ verifies their validity, including checks on the freshness of the timestamp ts and the correctness of the identity $x_i^{u,g}$. If all the received information is normal, $T_i^{u,g}$ blends all shares related to $x_i^{u,g}$ and reconstructs $f(x_i^{u,g})$ (Line 10). Then generates tags $tag_{i,l}^{u,g}$ for message authentication. Finally, $T_i^{u,g}$ packages the message into data report $msg_i^{u,g}$ and transmits it to the corresponding FN_g .

D. Data Aggregate

This protocol consists of two following algorithms: DataAggregate and FaultLocalization, as shown in **Protocol 4**.

1) DataAggregate: Upon receiving the data report $msg_i^{u,g}$ from $T_i^{u,g}$, FN_g first verifies its validity, including generating tags $tag_{i,l}^{u,g}$ with keys $k_{i,l}^{u,g}$ for verification. Next, FN_g chooses a random set of data terminals $I_{G_u^g} \sim (|I_{G_u^g}| = t_u^g)$. A polynomial $f(x)$ is then reconstructed using the pairs $(x_i^{u,g}, f(x_i^{u,g}))$ from the t_u^g terminals in the set, based on the Lagrange interpolation theorem (Line 3). As shown in (2)

$$\begin{aligned} f(x) &= \sum_{i=0}^{t_u^g-1} \left(f(x_i) \prod_{j=0, j \neq i}^{t_u^g-1} \frac{x - x_j}{x_i - x_j} \right) \bmod p^2 \\ &= \left(\sum_{i=1}^n r_{i,0} + \sum_{j=1}^m \left(x^j \sum_{i=1}^n c_{d_{i,j}^{u,g}} \right) + \sum_{j=1}^m (x^{m+j} \sum_{i=1}^n c_{s_{i,j}^{u,g}}) + \sum_{k=1}^{t_u^g-1-2m} \left(x^{2m+k} \sum_{i=1}^n r_{i,k} \right) + \sum_{i=1}^n a_{u,i}^g x^{t_u^g-1} \right) \bmod p^2 \end{aligned} \quad (2)$$

Here, $\sum_{i=1}^n c_{d_{i,j}^{u,g}}$ and $\sum_{i=1}^n c_{s_{i,j}^{u,g}}$ represent the aggregated ciphertexts of the j -th elements of the data vectors and selection vectors, respectively, from terminals in $I_{G_u^g}$. Then, FN_g extract $\sum_{i=1}^n c_{d_{i,1}^{u,g}}, \dots, \sum_{i=1}^n c_{d_{i,m}^{u,g}}, \sum_{i=1}^n c_{s_{i,1}^{u,g}}, \dots, \sum_{i=1}^n c_{s_{i,m}^{u,g}}$ and $\sum_{i=1}^n a_{u,i}^g$ from Polynomial $f(x)$. By verifying whether equation $a_{u,0}^g \stackrel{?}{=} \sum_{i=1}^n a_{u,i}^g$ holds, it can be determined whether

there are data terminals in G_u^g whose information has been tampered with, or whether there are lazy/malicious terminal nodes (Line 5). These data form a group G_u^g ciphertext vector $V_{G_u^g}$. Let FN_g -related groups be a set $\mathbb{G} = \{G_u^g | 1 \leq u \leq w, 1 \leq g \leq z\}$, then FN_g will compute aggregation V_{FN_g} , which includes $V_{G_u^g}$ from corresponding group, and generates tags tag_g for message authentication, then package the message into data report M_{FN_g} and send it to CS . The linear aggregation mechanism combined with zero-mean noise ensures that the proposed scheme preserves unbiasedness during both aggregation and reconstruction.

Let $\mathbb{F}_u^g$ denote the set of all shares collected from group G_u^g . If the equation $a_{u,0}^g \stackrel{?}{=} \sum_{i=1}^n a_{u,i}^g$ does not hold, FN_g will let $\mathbb{F}_u^g = \{f(x_i^{u,g}) | x_i^{u,g} \in G_u^g\}$ be the set of all shares and send it to KGC .

2) FaultLocalization: By removing the highest-degree term, the KGC reduces the received shares to a $(t - 2)$ -degree polynomial form (Line 2-4). Let $\mathbb{V}_u^g$ represent the set of valid shares, and $\mathbb{M}_u^g$ denote the set of faulty shares. Any subset of $(t - 1)$ honest shares reconstructs a valid polynomial (Line 6-9), while subsets containing faulty shares yield higher-degree polynomials, where $deg(\cdot)$ denotes the degree of the polynomial. By enumerating all such subsets, the KGC can deterministically identify the faulty sub-secret set $\mathbb{M}_u^g$ (Line 11-21). Finally, KGC broadcasts and blocks these data terminals.

In the worst-case scenario, the computational complexity of the FaultLocalization algorithm may be a combination of subsets, resulting in a complexity of $O(C(n, t_u^g))$. However, due to the random selection of subsets, the expected number of iterations in practice is usually small. Therefore, the average computational complexity can be approximated as $O(a \cdot t_u^{g2})$, where a is the expected number of experiments.

E. Result Report

This protocol only contains a ResReporting algorithm, as shown in **Protocol 5**.

Upon receiving the data report M_{FN_g} , CS first verifies its validity, including generating tags tag_g with keys k_g for verification. If the report is valid, CS aggregates the ciphertext data received from all relevant fog nodes to compute V_{CS} (Line 3). As shown in (3)

$$\begin{aligned} V_{CS} &= \sum_{FN_g \in \mathbb{F}} V_{FN_g} \\ &= \left(\sum_{j=1}^f c_{d_{j,1}^{u,g}}, \dots, \sum_{j=1}^f c_{d_{j,m}^{u,g}}, \sum_{j=1}^f c_{s_{j,1}^{u,g}}, \dots, \sum_{j=1}^f c_{s_{j,m}^{u,g}} \right) \\ &= (C_{d_1}, \dots, C_{d_m}, C_{s_1}, \dots, C_{s_m}) \end{aligned} \quad (3)$$

Let CS -related fog nodes be a set $\mathbb{F} = \{FN_g | 1 \leq g \leq z\}$. It then proceeds to decrypt the aggregated ciphertext to obtain the plaintext aggregated values $\{M_1, \dots, M_m\}$ and the corresponding selected data $\{S_1, \dots, S_m\}$ (Line 5-6). These plaintext results can be statistically analyzed to compute the mean value $\{Avg(M_1), \dots, Avg(M_m)\}$ (Line 7).

In practice, these averages are commonly used to characterize the overall health status of a group, such as the average heart rate,

Protocol 4: Data Aggregate.

$\text{DataAggregate}(I_{G_u^g}, y_g, a_{u,0}^g, msg_i^{u,g}) \rightarrow (V_{G_u^g}, V_{FN_g}, M_{FN_g}, tag_g)$

- 1: FN_g extracts $f(x_i^{u,g})$, $x_i^{u,g}$ and ts from data report $msg_i^{u,g}$.
- 2: Verifies timestamp ts and identity $x_i^{u,g}$.
- 3: Computes $f(x) = \sum_{i=0}^{t_u^g-1} (f(x_i) \prod_{j=0, j \neq i} \frac{x-x_j}{x_i-x_j}) \bmod p^2$
- 4: Extracts coefficient $\sum_{i=1}^n c_{d_{i,1}}^{u,g}, \dots, \sum_{i=1}^n c_{d_{i,m}}^{u,g}, \sum_{i=1}^n c_{s_{i,1}}^{u,g}, \dots, \sum_{i=1}^n c_{s_{i,m}}^{u,g}$ and $\sum_{i=1}^n a_{u,i}^g$ of x in $f(x)$.
- 5: **if** $a_{u,0}^g = \sum_{i=1}^n a_{u,i}^g$ **then**
- 6: Sets aggregation values of ciphertext in j -dimension $V_{G_u^g} \leftarrow (\sum_{i=1}^n c_{d_{i,1}}^{u,g}, \dots, \sum_{i=1}^n c_{d_{i,m}}^{u,g}, \sum_{i=1}^n c_{s_{i,1}}^{u,g}, \dots, \sum_{i=1}^n c_{s_{i,m}}^{u,g})$
- 7: Computes aggregate values for each group $V_{FN_g} \leftarrow \sum_{G_u^g \in \mathbb{G}} V_{G_u^g}$
- 8: Computes $tag_g \leftarrow MAC_{k_g}(y_i || ts || V_{FN_g})$
- 9: Generates $M_{FN_g} \leftarrow y_i || ts || V_{FN_g} || tag_g$
- 10: Sends data report M_{FN_g} to CS
- 11: **else**
- 12: Sends $\mathbb{F}_u^g = \{f(x_i^{u,g}) | x_i^{u,g} \in G_u^g\}$ to KGC .
- 13: **end if**
- 14: **return**

$\text{FaultLocalization}(\mathbb{F}_u^g, \mathbb{G}, t_u^g) \rightarrow (\mathbb{V}_u^g, \mathbb{M}_u^g)$

- 1: KGC received and parsed $\mathbb{F}_u^g$.
- 2: **for each** $f(x_i^{u,g})$ in $\mathbb{F}_u^g$ **do**
- 3: Computes $f(x_i^{u,g})' \leftarrow (f(x_i^{u,g}) - a_{u,0}^g x^{t_u^g-1}) \bmod p^2$
- 4: **end for**
- 5: Let $\mathbb{V}_u^g \leftarrow \emptyset$ and $\mathbb{M}_u^g \leftarrow \emptyset$
- 6: **while** $\deg(f(\cdot)) \neq t_u^g - 2$ **do**
- 7: Randomly select $t_u^g - 1$ shares in $\mathbb{F}_u^g$ as subset $\mathbb{I}_u^g$
- 8: Computes $f(x) \leftarrow \sum_{x_i \in \mathbb{I}_u^g} (f(x_i)' \prod_{j=0, j \neq i} \frac{x-x_j}{x_i-x_j}) \bmod p^2$
- 9: **end while**
- 10: Let $\mathbb{V}_u^g \leftarrow \mathbb{I}_u^g$
- 11: **while** shares in $\mathbb{F}_u^g$ have not been fully traversed **do**
- 12: $\mathbb{I}_u^g$ randomly remove a share with $|\mathbb{I}_u^g| = t_u^g - 2$
- 13: Randomly select and mark $f(x_i^{u,g})' \in \mathbb{F}_u^g \setminus \mathbb{V}_u^g$ as tep
- 14: Add tep to $\mathbb{I}_u^g$
- 15: Computes $f(x) \leftarrow \sum_{x_i \in \mathbb{I}_u^g} (f(x_i)' \prod_{j=0, j \neq i} \frac{x-x_j}{x_i-x_j}) \bmod p^2$
- 16: **if** $\deg(f(\cdot)) = t_u^g - 2$ **then**
- 17: $\mathbb{V}_u^g \leftarrow \mathbb{I}_u^g$
- 18: **else**
- 19: $tep \in \mathbb{M}_u^g$
- 20: **end if**
- 21: **end while**
- 22: Broadcast and block $\mathbb{M}_u^g$ in the system.
- 23: **return**

Protocol 5: Result Report.

$\text{ResReporting}(M_{FN_g}, SK_j, SK_{m+j}) \rightarrow (M_j, S_j, Avg(M_j))$

- 1: CS extract V_{FN_g} , y_g and ts from data report M_{FN_g} .
- 2: Verify timestamp ts and identity y_g .
- 3: Computes aggregation values set of ciphertext $V_{CS} = \sum_{FN_g \in \mathbb{F}} V_{FN_g}$ using (3)
- 4: **for** $j = 1, 2, \dots, m$ **do**
- 5: Computes aggregation values in j -dimension $M_j = (C_{d_j} - SK_j) \bmod \mathbb{N}$
- 6: Computes aggregation values of selected data in j -dimension $S_j = (C_{s_j} - SK_{m+j}) \bmod \mathbb{N}$
- 7: Computes mean value $Avg(M_j) = \frac{M_j}{S_j}$.
- 8: **end for**
- 9: **return** $M_j, S_j, Avg(M_j)$ ($1 \leq j \leq m$).

blood pressure, or other physiological indicators. For example, in community-level influenza surveillance, variations in the average body temperature across different time periods can be used to estimate influenza incidence. Similarly, in pharmaceutical studies, the average values of physiological indicators measured at different time points after drug administration can serve as effective metrics for evaluating and predicting drug efficacy.

V. THEORETICAL ANALYSIS

A. Correctness

Theorem 1: Only when $a_{u,0}^g$ in **Protocol 4** is correctly verified, can the information submitted by the data terminal be correct.

Proof: After receiving the message from $T_l^{u,g}, T_i^{u,g}$ superimposes all polynomials related to $x_i^{u,g}$ to obtain $f(x_i^{u,g})$, as shown in (4), shown at the bottom of next page. The $a_{u,i}^g$ distributed by KGC to each terminal is embedded in a polynomial, just like the ciphertext. After superposition, each subsecret contains $\sum_{i=1}^n a_{u,i}^g$. $\square$

Theorem 2: Only after decryption by CS can the aggregated data of each dimension of the user be obtained.

Proof: Decrypts $(C_{d_1}, \dots, C_{d_m}, C_{s_1}, \dots, C_{s_m})$ to obtain M_j and S_j , as shown in the following equation:

$$\begin{aligned}
 M_j &= (C_{d_j} - SK_j) \bmod \mathbb{N} \\
 &= \left(\sum_{i=1}^f c_{d_{i,j}}^{u,g} - \sum_{i=1}^f v_{i,j}^{u,g} \right) \bmod \mathbb{N} \\
 &= \left((c_{d_{1,j}}^{u,g} - v_{1,j}^{u,g}) + \dots + (c_{d_{f,j}}^{u,g} - v_{f,j}^{u,g}) \right) \bmod \mathbb{N} \\
 &= \sum_{i=1}^f (s_{i,j}^{u,g} md_{i,j}^{u,g} + v_{i,j}^{u,g} - v_{i,j}^{u,g}) \bmod \mathbb{N} \\
 &= (s_{1,j}^{u,g} md_{1,j}^{u,g} + s_{2,j}^{u,g} md_{2,j}^{u,g} + \dots + s_{f,j}^{u,g} md_{f,j}^{u,g}) \bmod \mathbb{N} \\
 &= \sum_{i=1}^f s_{i,j}^{u,g} md_{i,j}^{u,g}
 \end{aligned}$$

Therefore, the decrypted result is the product of the selected data $s_{i,j}^{u,g}$ and the DP-perturbed data $md_{i,j}^{u,g}$.

$$\begin{aligned}
S_j &= (C_{s_j} - SK_{m+j}) \bmod \mathbb{N} \\
&= \left(\sum_{i=1}^f c_{s_{i,j}}^{u,g} - \sum_{i=1}^f v_{i,m+j}^{u,g} \right) \bmod \mathbb{N} \\
&= \left((c_{s_{1,j}}^{u,g} - v_{1,m+j}^{u,g}) + \cdots + (c_{s_{f,j}}^{u,g} - v_{f,m+j}^{u,g}) \right) \bmod \mathbb{N} \\
&= \sum_{i=1}^f (s_{1,j}^{u,g} + v_{1,m+j}^{u,g} - v_{1,m+j}^{u,g}) \bmod \mathbb{N} \\
&= (s_{1,j}^{u,g} + s_{2,j}^{u,g} + \cdots + s_{f,j}^{u,g}) \bmod \mathbb{N} \\
&= \sum_{i=1}^f s_{i,j}^{u,g}
\end{aligned}$$

Finally, the decrypted result is the selection of aggregated data $\sum_{i=1}^f s_{i,j}^{u,g}$, and the theorem is proved. $\square$

B. Data Utility

The effectiveness of differential privacy technology in privacy preserving data aggregation mainly depends on the relative error of the aggregation results. Therefore, in order to evaluate and compare the effectiveness of differential privacy protection data aggregation schemes with different privacy parameters, we conducted experiments to compare the relative error size performance under different parameters.

We chose a dataset containing four hundred and fifty thousand synthetic but real health data sets to predict the health status of patients. The experiments were conducted using the Enhanced Large Health Dataset [20] obtained from a public repository. This dataset was created using real data monitored by medical sensors. We randomly selected heart rate data from 1 to 10000 users from a total of 450000 medical sensor datasets, added noise to each and summarized them, and finally calculated the percentage of error. We compared the percentage of error caused by λ values of 0.5 and 1.0 with the original data. The privacy

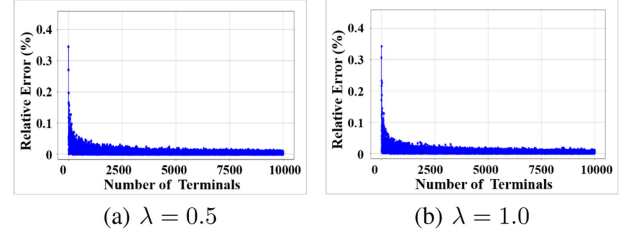


Fig. 2. Relative error caused by various privacy parameters λ .

budgets $\lambda = 0.5$ and $\lambda = 1$ are chosen as representative settings in our experiments. These values correspond to moderate and relatively relaxed privacy levels, respectively, and have been extensively used in prior differential privacy study [21]. This choice enables a clear comparison of the privacy–utility trade-off under different noise intensities. As shown in Fig. 2.

C. Privacy Protection

1) *Confidentiality*: Assuming there is an external attacker intercepting, altering, monitoring, or even fabricating information between the communicating parties. This scheme performs simple encryption processing on personal data. The key $v_{i,j}^{u,g}$ is random and one-time. The attackers need to find plaintext from the generated ciphertext with a probability of $\frac{1}{\mathbb{N}}$ even if they obtain $md_{i,j}^{u,g}$, while attackers need a probability of $\frac{1}{p^{(t_u^g-1)}}$ to crack ciphertext from polynomials. Therefore, the probability of obtaining sensitive data is $P = \frac{1}{\mathbb{N} \cdot p^{(t_u^g-1)}}$. If let $|\mathbb{N}| = 101$, $t_u^g = 10$, then P is a very small number, so the probability is impossible.

2) *Resist Differential Attack*: According to the threat model in **Section IV**, attackers may obtain personal user data through differential attacks. Our proposed scheme can ensure differential privacy and avoid the leakage of fine-grained private data. On the one hand, utilizing differential privacy techniques and the infinite divisibility of the Laplacian distribution, partial noise is extracted from the gamma distribution. Formally, a Laplace distributed random variable $L \sim \text{lap}(\theta, \lambda)$ can be simulated using the sum of $2n$ random variables.

$$\begin{aligned}
f(x_i^{u,g}) &= \sum_{l=1}^n f_l(x_i^{u,g}) \\
&= \left(r_{1,0}^{u,g} + \sum_{j=1}^m c_{d_{1,j}}^{u,g} (x_i^{u,g})^j + \sum_{j=1}^m c_{s_{1,j}}^{u,g} (x_i^{u,g})^{m+j} + a_{u,1}^g (x_i^{u,g})^{2m+1} + \sum_{k=1}^{t_u^g-2-2m} r_{1,k}^{u,g} (x_i^{u,g})^{2m+1+k} + \cdots + r_{n,0}^{u,g} \right. \\
&\quad \left. + \sum_{j=1}^m c_{d_{n,j}}^{u,g} (x_i^{u,g})^j + \sum_{j=1}^m c_{s_{n,j}}^{u,g} (x_i^{u,g})^{m+j} + a_{u,n}^g (x_i^{u,g})^{2m+1} + \sum_{k=1}^{t_u^g-2-2m} r_{n,k}^{u,g} (x_i^{u,g})^{2m+1+k} \right) \bmod p^2 \\
&= \left(\sum_{i=1}^n r_{i,0}^{u,g} + \sum_{j=1}^m ((x_i^{u,g})^j \sum_{i=1}^n c_{d_{i,j}}^{u,g}) + \sum_{j=1}^m ((x_i^{u,g})^{m+j} \sum_{i=1}^n c_{s_{i,j}}^{u,g}) + \sum_{i=1}^n a_{u,i}^g (x_i^{u,g})^{2m+1} + \sum_{k=1}^{t_u^g-2-2m} ((x_i^{u,g})^{2m+1+k} \right. \\
&\quad \left. \sum_{i=1}^n r_{i,k}^{u,g}) \right) \bmod p^2
\end{aligned} \tag{4}$$

On the other hand, we have demonstrated that aggregated data in the j -th dimension, combined with simulated noise extracted from the Laplacian distribution, can also achieve ϵ -differential privacy. For example, suppose the attacker obtains two perturbed aggregated data $A(D_u) = D_u + L_u$ and $A(D_v) = D_v + L_v$, where D_u and D_v are two adjacent aggregates, L_u and L_v are the corresponding noise from the Laplace distribution, with a scale of $\lambda = \frac{\Delta_Q}{\epsilon}$. Therefore, we have the following equation

$$\begin{aligned} \frac{Pr[A(D_u) = x]}{Pr[A(D_v) = x]} &= \frac{\frac{1}{2\lambda} \exp\left(-\frac{|x - A(D_u)|}{\lambda}\right)}{\frac{1}{2\lambda} \exp\left(-\frac{|x - A(D_v)|}{\lambda}\right)} \\ &= \exp\left(\frac{|A(D_u) - A(D_v)|}{\lambda}\right) \\ &\leq \exp\left(\frac{\Delta_A}{\lambda}\right) = \exp(\epsilon) \end{aligned}$$

According to the differential privacy axiom proposed in [22], differential invariance is defined as a property of data transformation, which means that within the framework of privacy protection, the transformation process of data must be able to maintain the privacy of the original data, that is, the transformed data should not disclose any additional information about the individual. Converting the perturbed personal privacy data into ciphertext C_{d_j} can also satisfy ϵ -differential privacy.

In addition, we consider each dimension of personal data as an independent set, and there is no intersection between personal data in different sets, satisfying the parallel composability of differential privacy in **Section IV**. The perturbed aggregation results of the m -dimensional data after adding noise satisfies $\max\{\epsilon_1, \epsilon_2, \dots, \epsilon_m\}$ -differential privacy.

3) *Resist Replay Attack*: Assuming an external attacker captures legitimate communication data and resends it to the target system, attempting to bypass authentication, authorization, or perform incorrect operations. In this scheme, add timestamps ts to communication data msg to ensure that each request is made based on a specific time. CS and FN_g can verify whether the timestamp of the request is within the allowed time window. If the timestamp of the request is too old, the server will reject the request.

D. Data Integrity

Add a verification mechanism at the fog node, which can effectively detect and verify the existence of lazy or malicious terminal nodes by verifying $a_{u,0}^g \stackrel{?}{=} \sum_{i=1}^n a_{u,i}^g$, thereby ensuring the overall performance and data processing efficiency of the system. If there is a lazy/malicious terminal that has not submitted/modified its data, then the equation does not hold.

E. Fault-Tolerant

In practical applications, terminals are prone to malfunctions, and it is unrealistic to require all terminals to execute protocols correctly and without errors. This scheme has fault-tolerant properties, allowing no more than $(n - t_u^g)$ terminals in a group to fail. For example, when a timeout occurs, each terminal only

TABLE II
FUNCTIONALITY COMPARISONS OF EXISTING SCHEMES

Schemes	Primitives	DC	DI	RC	RLN	OD	FT
Guo et al. [4]	HE	✓	✓	×	✓	×	×
VMEMDA [3]	HE	✓	✓	×	×	×	×
MDA-FLH [5]	SS+HE	✓	✓	✓	×	×	✓
OPERA [7]	SS	✓	✓	✓	×	✓	✓
Peng et al. [11]	HE	✓	✓	×	×	×	✓
LPPMM-DA [12]	HE	✓	✓	✓	×	×	×
SVODA (Ours)	SS+DP	✓	✓	✓	✓	✓	✓

receives data from the other $(n - 1)$ terminals and still reports mixed messages to the fog node. Due to the presence of sufficient t_u^g shared values to assist fog nodes in recovering aggregated Lagrangian polynomials, even if $(n - t_u^g)$ terminals fail, it will not affect the subsequent execution of the protocol.

F. Collusion Prevention

Assuming the existence of internal attackers, we will describe three scenarios as follows:

If CS and FN_g collude, CS might share the key SK_j and SK_{m+j} with the fog node. However, since the terminal only sends aggregated shared data to the fog node, the fog node can only access the aggregated data for each group, without being able to retrieve individual data. This effectively protects the privacy of the terminal users. Let C_x be the aggregated data item of n terminals. For fog nodes, the probability of detecting a private message is $P = \frac{(C_x - n)!}{N \cdot C_x!}$. When $|\mathbb{N}| = 101$, $t_u^g = 10$, this probability is also almost impossible.

If CS and $T_i^{u,g}$ collude, CS might share the key SK_j and SK_{m+j} with data terminals. After decryption, the terminal $T_i^{u,g}$ can obtain the final aggregated result, but the terminal $T_i^{u,g}$ cannot infer any other terminal's $T_l^{u,g} \sim (1 \leq l \leq f, \sim l \neq i)$ data from individual data and aggregated result data.

If FN_g and $T_i^{u,g}$ collude, $T_i^{u,g}$ might share the data report $msg_i^{u,g}$ with fog node, according to the principle of secret sharing threshold for secret recovery, as long as the fog node conspires with less than t_u^g terminals, the fog node cannot recover the secret and obtain the aggregated value.

VI. PERFORMANCE EVALUATION

A. Functionality Comparisons

We conducted a comprehensive comparison between the proposed scheme and several representative data aggregation schemes in terms of their supported functionalities. The evaluation criteria are divided into two main categories: security and flexibility. The security aspects include Primitives, Data Confidentiality (DC), Data Integrity (DI), Resistance to Collusion (RC), and Resistance to Lazy Nodes (RLN). The flexibility aspects encompass Optional Dimensions (OD) and Fault Tolerance (FT). A summary of the functionality comparisons based on these criteria is presented in Table II.

Guo et al.'s scheme [4] utilized an improved symmetric homomorphic encryption method to ensure data confidentiality

and integrity. However, it lacks support for fault tolerance, differential privacy, and other advanced functionalities. Similarly, VMEMDA [3] enhances the Paillier cryptosystem by introducing super-increasing sequences to achieve multidimensional data aggregation in WBANs. MDA-FLH [5] combined the Chinese Remainder Theorem (CRT) with secret sharing to enable fine-grained linear homomorphic aggregation, supporting fault tolerance and collusion resistance. However, it does not offer features such as optional dimensions, differential privacy, and resistance to lazy nodes. OPERA [7] integrated AES encryption with secret sharing to enable optional dimension aggregation and fault tolerance, as well as resistance to collusion. Nevertheless, it does not provide support for resistance to lazy nodes and differential privacy. Peng et al.'s scheme [11], based on CRT and homomorphic encryption, achieved multidimensional aggregation in WBANs, yet lacks support for most of the evaluated functionalities. LPPMM-DA [12] employed ring signatures to resist collusion in multi-subset aggregation scenarios but fails to meet requirements such as fault tolerance, differential privacy, or support for optional dimensions.

In contrast, our proposed scheme supports all seven key functionalities, effectively addressing the limitations found in existing works. It ensures both security and flexibility, demonstrating its comprehensive and practical applicability in real-world WBANs scenarios.

B. Experimental Setup

The comparative experiments were conducted on a laptop running Windows 11 (64-bit) with 16.0 GB RAM and an AMD Ryzen 5 5600 U processor operating at 2.30 GHz with integrated Radeon Graphics. All implementations were developed in Python, employing the Secret Sharing library and the Pairing-Based Cryptography (PBC) library.

A key design objective of the proposed scheme is to reduce the computational burden on each participating entity in the data aggregation process, thereby improving overall system efficiency. To demonstrate this, we conducted a series of comparative experiments. First, the computational overhead of the proposed scheme was evaluated and compared with that of Guo et al.'s scheme [4], OPERA [7], and LPPMM-DA [12]. The evaluation mainly focused on three types of entities: data terminals, fog nodes, and cloud servers. Next, we assessed communication and storage overhead through comparative analysis. In the aggregation system, we assume n users, m data dimensions, and z aggregators. For consistency, we set $z = 10$ and ensured that all schemes were evaluated under the same security level. Considering the characteristics of WBANs, 20 data dimensions are generally sufficient. Accordingly, the underlying polynomial operations were configured with an order of 50 for performance evaluation.

C. Computation Cost Comparison

Before evaluating the computational cost of our scheme, we need to consider some of the operations involved in the aggregation process. To simplify the notation, we have introduced some symbols to represent these operations, as shown in Table III.

TABLE III
NOTATIONS OF EXECUTION TIME

Notations	Descriptions	Test Results
T_{poly}	Time for generating polynomial	0.221 ms
T_{Rep}	Time for recovering polynomial	6.024 ms
T_{dp}	Time for differential privacy	0.048 ms
T_{ez}	Time for exponentiation in $\mathbb{Z}_N$	0.147 ms
T_m	Time for multiplication in $\mathbb{Z}_N$	0.076 ms
T_{mod}	Time for modular $\mathbb{N}$ operation	0.001 ms
T_{modN^2}	Time for modular $\mathbb{N}^2$ operation	0.004 ms
T_{pair}	Time for a bilinear pairing	1.936 ms
T_{Hp}	Time for Hash to point operation	0.112 ms
T_{AES_e}	Time for AES-128 encryption	0.135 ms
T_{AES_d}	Time for AES-128 decryption	0.054 ms
T_{ECC}	Time for a point multiplication in ECC	0.827 ms
$ p $	The size of large prime	512 bits
$ q $	The size of the order q of cyclic group on elliptic curve	160 bits
$ ID_s $	The size of identity	32 bits
$ T_s $	The size of the timestamp	64 bits
$ \mathbb{G}_1 $	The size of element in $\mathbb{G}_1$	512 bits
$ \mathbb{G}_2 $	The size of element in $\mathbb{G}_2$	1024 bits

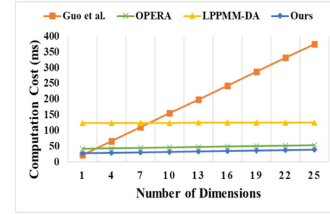


Fig. 3. Computation cost on data terminals ($n=100$).

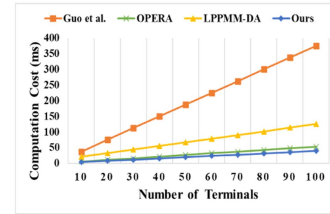


Fig. 4. Computation cost on data terminals ($m=25$).

It should be noted that since the cost of hash operations is relatively small and can be ignored, the computational cost of hash operations has not been considered.

1) *Computation Cost on Data Terminals:* Figs. 3, 4, and Table IV present the computation costs on data terminals with respect to the data dimension and the number of terminals. When the number of terminals is 100 and the data dimension is 25, the computational costs of our scheme (SVODA), Guo et al. [4], OPERA [7], and LPPMM-DA [12] are 39.40 ms, 375.10 ms, 53.45 ms, and 125.49 ms, respectively. Overall, our scheme consistently outperforms the comparative approaches, demonstrating substantially lower computational overhead as the number of terminals and data dimensions increase.

The superior performance of the proposed scheme on data terminals is primarily attributed to its lightweight design. Unlike Guo et al. [4], OPERA [7], and LPPMM-DA [12], which rely on computationally expensive asymmetric homomorphic

TABLE IV
THE COMPARISONS OF COMPUTATION COST BETWEEN EXISTING SCHEMES AND OURS

Schemes	Data Terminal Layer	Fog Server Layer	Cloud Server Layer
Guo et al. scheme [4]	$nmT_{ez} + nT_m$	$nT_{pair} + 2zT_{pair} + zT_m$	$(z + 2)T_{pair} + nmT_{ez}$
OPERA [7]	$n(T_{AES_e} + T_{poly}) + (n - 1)T_{AES_d}$ $+ nm(T_{mod} + T_{modN^2})$	$nzT_{AES_d} + zT_{Rep} + 2mzT_{AES_e}$	$2mzT_{AES_d} + 2mT_{mod}$
LPPMM-DA [12]	$mT_m + (n + 8)T_{ECC} + (3n + 6)T_{Hp}$	$(2n + 1)zT_{ECC} + (n + 1)zT_{Hp}$	$(2n + 3)T_{ECC} + (2n + 1)T_{Hp}$ $+ (m + 5)T_m$
SVODA (Ours)	$nT_{poly} + nT_{dp} + nmT_{mod} + nmT_{modN^2}$	zT_{Rep}	$2mT_{mod}$

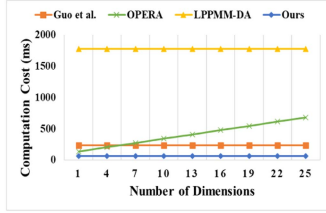


Fig. 5. Computation cost on fog nodes ($n=100$).

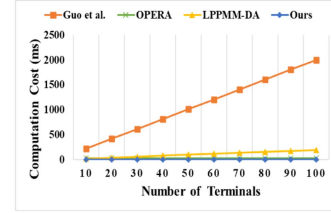


Fig. 7. Computation cost on cloud servers ($m=25$).

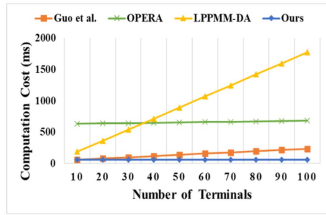


Fig. 6. Computation cost on fog nodes ($m=25$).

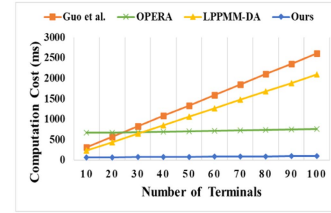


Fig. 8. Total computation cost ($m=25$).

encryption or AES-based symmetric encryption, our scheme eliminates these overheads. Instead, it integrates differential privacy mechanisms with low-complexity operations to ensure privacy preservation. This design choice significantly reduces the computational burden on data terminals, making the scheme particularly well-suited for resource-constrained wearable devices in WBANs.

2) *Computation Cost on Fog Nodes:* Figs. 5, 6, and Table IV present the computation costs at fog nodes with respect to the data dimension and the number of terminals. When the number of terminals is 100 and the data dimension is 25, the computational costs of our scheme (SVODA), Guo et al. [4], OPERA [7], and LPPMM-DA [12] are 60.24 ms, 233.08 ms, 681.71 ms, and 1775.39 ms. Overall, the proposed scheme demonstrates significantly lower computational overhead at fog nodes compared with the baseline schemes, particularly as the number of terminals and data dimensions increase.

The performance advantage is mainly due to the lightweight operations required at the fog node. Specifically, after receiving the data reports, the fog node performs a single SS recovery operation, instead of executing computationally intensive homomorphic cryptographic operations as in the comparative schemes. This design effectively reduces processing costs and enhances the scalability of the aggregation system.

3) *Computation Cost on Cloud Servers:* Fig. 7 and Table IV illustrates the computational overhead on the cloud server as the number of terminals increases. With 100 terminals and 25-dimensional data, the computational costs of our scheme (SVODA), Guo et al. [4], OPERA [7], and LPPMM-DA [12] are 0.05 ms, 2003.32 ms, 27.05 ms, and 192.67 ms, respectively. Clearly, the proposed scheme imposes a substantially lower computational load on the cloud server compared with the baseline schemes, and this advantage becomes more pronounced as the number of terminals grows.

The efficiency gain primarily arises from the lightweight decryption design. Specifically, decryption in our scheme requires only $2m$ modular operations, while the comparative schemes rely on computationally expensive symmetric or asymmetric decryption. This design significantly reduces the computational burden at the cloud server, thereby enhancing scalability and overall system efficiency.

4) *Total Computation Cost:* Finally, we compared the total computational cost of the system. As shown in Fig. 8, when the number of terminals is 100 and the data dimension is 25, the total computational costs of our scheme (SVODA), Guo et al. [4], OPERA [7], and LPPMM-DA [12] are 99.69 ms, 2611.41 ms, 762.24 ms, and 2093.55 ms, respectively. The results demonstrate that the total computational cost of our scheme consistently remains lower than that of the comparative

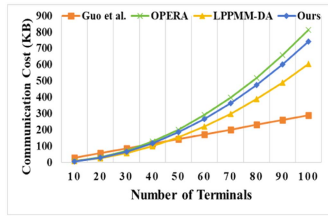


Fig. 9. Communication cost of T-to-FN.

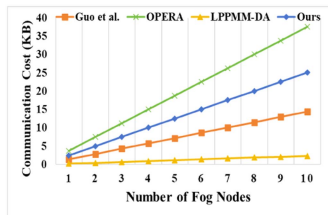


Fig. 10. Communication cost of FN-to-CS.

schemes as the number of terminals increases, underscoring its efficiency advantage.

As the number of terminals increases, the overall computational overhead remains relatively low, since the proposed scheme does not involve costly symmetric and asymmetric cryptographic operations, effectively reducing the computational load of each system entity and achieving a lightweight design. This reduction in aggregation overhead is particularly beneficial in real-world WBANs, where it can extend sensor lifespan, improve user comfort, and enhance clinical reliability.

D. Communication Cost Comparison

Before evaluating the communication cost, we first define the data sizes involved in the operations, as summarized in Table III. Since differential privacy operations have negligible impact on data size, the data before and after applying differential privacy is considered identical.

1) *Communication Cost of T-to-FN*: As shown in Fig. 9, we compare the communication overhead of the proposed scheme (SVODA) with existing schemes at the terminal level. The results show that SVODA incurs the similar communication overhead as OPERA [7], but higher than Guo et al.'s scheme [4] and LPPMM-DA [12]. Specifically, when the data dimension reaches 20, the communication costs of SVODA, Guo et al.'s scheme, OPERA, and LPPMM-DA are 742.19 KB, 289.06 KB, 814.45 KB, and 603.91 KB, respectively. The relatively higher overhead of SVODA is attributed to the construction of sub-secrets in the secret sharing process, which requires terminal interaction and scales with the number of terminals. Nonetheless, this trade-off yields significantly lower computational costs compared to Guo et al.'s scheme and LPPMM-DA, making our scheme (SVODA) more suitable for WBAN applications.

2) *Communication Cost of FN-to-CS*: As shown in Fig. 10, we compare the communication overhead of the proposed scheme with existing schemes at the fog node. The results show that our scheme achieves better communication efficiency than

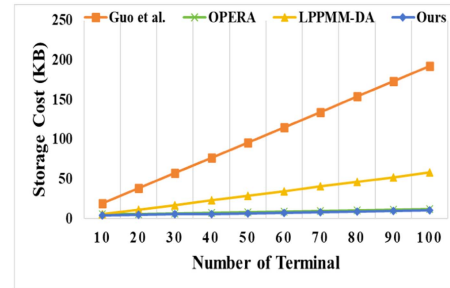


Fig. 11. Storage cost on data terminals.

OPERA [7], but incurs higher costs than Guo et al.'s scheme [4] and LPPMM-DA [12]. Specifically, when the data dimension reaches 20, the communication overheads of our scheme (SVODA), Guo et al.'s scheme, OPERA, and LPPMM-DA are 25.12 KB, 14.45 KB, 37.58 KB, and 2.38 KB, respectively, on FN-to-CS communication. The relatively higher overhead of our scheme arises because the fog node must collect data, timestamps, and identities from each terminal, which together exceed the size of the encrypted data. Nevertheless, this design enables improved computational efficiency compared to Guo et al.'s scheme, OPERA, and LPPMM-DA, highlighting the trade-off between communication and computation costs.

E. Storage Cost Comparison

Before evaluating the storage overhead, we first define the data sizes involved in the operations, as summarized in Table III. It is noted that the hash function adopted in our scheme is SHA-256.

1) *Storage Cost on Data Terminals*: As shown in Fig. 11, we compare the storage overhead of data terminals during Data Blend stage under varying numbers of terminals. When the number of terminals is 100, the storage overhead of our scheme (SVODA), Guo et al.'s scheme [4], OPERA [7], and LPPMM-DA [12] is 10.73 KB, 192.19 KB, 12.23 KB, and 58.20 KB, respectively. The results show that the storage overhead of our scheme grows slowly with the number of terminals and remains consistently lower than that of Guo et al.'s scheme and LPPMM-DA.

The advantage in storage efficiency primarily stems from the lightweight design of our scheme. Unlike Guo et al.'s scheme and LPPMM-DA, which require additional public-private key pairs for asymmetric HE or symmetric encryption keys, our approach avoids such key management overhead. Instead, it integrates differential privacy to minimize storage consumption at the terminal side. This design makes the scheme more suitable for resource-constrained devices in WBAN scenarios, where storage efficiency directly affects system practicality.

2) *Storage Cost on Fog Nodes*: As illustrated in Fig. 12, we evaluate the storage cost on fog nodes under varying numbers of terminals. The results demonstrate that when the number of terminals is 100, the storage overhead of our scheme (SVODA), Guo et al.'s scheme [4], OPERA [7], and LPPMM-DA [12] is 14.13 KB, 193.92 KB, 14.14 KB, and 47.66 KB. Overall, our scheme achieves comparable storage consumption to OPERA while maintaining a significantly lower cost than Guo et al.'s scheme and LPPMM-DA.

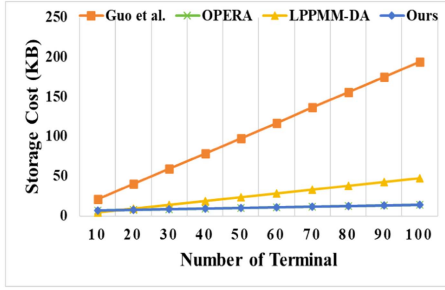


Fig. 12. Storage cost on fog nodes.

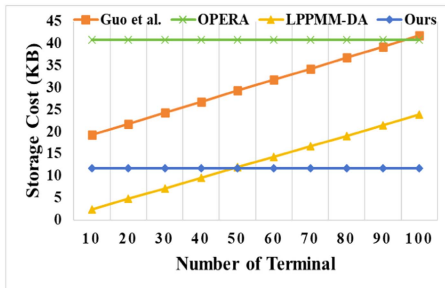


Fig. 13. Storage cost on cloud servers.

The analysis further reveals that the storage overhead of our scheme grows slowly as the number of terminals increases, with a growth rate notably lower than that of the comparative schemes. This advantage stems from the fact that our design eliminates the need to store multiple signatures for data integrity verification at the fog node. Instead, a lightweight verification mechanism is adopted, thereby reducing storage consumption. Consequently, the proposed architecture demonstrates superior scalability and efficiency in fog node.

3) *Storage Cost on a Cloud Servers*: As illustrated in Fig. 13, we evaluate the storage overhead of cloud servers with varying numbers of terminals. When the number of terminals is 100, the storage overhead of our scheme (SVODA), Guo et al.'s scheme [4], OPERA [7], and LPPMM-DA [12] is 11.64 KB, 41.76 KB, 40.84 KB, and 23.83 KB. Overall, our scheme consistently achieves the lowest storage cost among all schemes, with a particularly notable reduction compared to Guo et al.'s scheme and OPERA.

Further analysis shows that the storage overhead of our scheme remains constant as the number of terminals increases, while the comparative schemes exhibit steady growth. This is because our scheme avoids storing decryption private keys required by asymmetric HE and eliminates the need to store public keys for signature verification on cloud server. As a result, the proposed design offers clear advantages in scalability and efficiency with respect to cloud server storage overhead.

VII. RELATED WORKS

With the proliferation of wearable devices and IoT-based healthcare systems, WBANs generate and aggregate large volumes of sensitive health data. Existing studies highlight

that while data aggregation improves efficiency and decision-making, it also introduces serious concerns of data leakage and misuse [23]. To address these challenges, research efforts have mainly focused on two directions: **Cryptography-based Schemes** and **DP-based Schemes**.

Cryptography-based Schemes: Many methods for privacy preserving data aggregation have been proposed. For example, Peng et al. [24] constructed a linear homomorphic encryption algorithm using the Chinese Remainder Theorem to achieve privacy preserving data aggregation, after decrypting the aggregated data, the extracted dimensional data is the result of linear operations. There are also some data aggregation schemes [19], [25] based on homomorphic encryption algorithm, data can be effectively aggregated to achieve multi-dimensional data aggregation, but it cannot avoid malicious control centers accessing the original data. In traditional data aggregation methods, centralized architecture poses a risk of single point of failure or centralized attacks. By applying secret sharing [5], [7], each data fragment can be distributed across multiple locations for computation and processing, improving fault tolerance and reducing the risk of data leakage.

By applying secret sharing technology, data fragments can be distributed across multiple locations for computation and processing, thereby improving fault tolerance and reducing the risk of data leakage. Alghamdi et al. [14] proposed an end-to-end data aggregation scheme based on SS. Yang et al. [26] proposed the encryption scheme based on lattice cryptography, offering a triple-layered assurance for the secure implementation of SS. Wagh et al. [27] proposed a framework based on secret sharing to prevent single point of failure. Vassantlal [28] proposed a protocol stack for dynamic proactive secret sharing that allows implementing confidentiality in Byzantine Fault-Tolerant state machine replication systems. Zhang et al. [29] proposed an identity based aggregate signature scheme that combines BGN public key cryptography and secret sharing to achieve fault tolerance. Chen et al. [5] proposed an MDA-FLH multidimensional data aggregation scheme with fine-grained linear homomorphism, combined with the Chinese remainder theorem and Paillier encryption, using extended secret sharing to achieve fault tolerance. Although these schemes effectively ensure data security and fault tolerance, the time complexity of public key encryption is high, and the computational burden may become a bottleneck for the system.

DP-based Schemes: Differential privacy technology has low computational overhead and protects data privacy by introducing noise to prevent attackers from obtaining individual information from data analysis. Wang et al. [30] proposed an additive mechanism for localized differential privacy, which enhances utility and robustness in the data aggregation process. He et al. [31] utilized differential privacy to achieve accurate calculation of data aggregation results in smart grid scenarios without compromising its security. Gai et al. [32] proposed a smart grid data aggregation scheme based on local differential privacy, which is free from the intervention of trusted third parties. Zheng et al. [33] proposed a differential privacy computing mechanism for smart grids, which extends the time domain of data. Han et al. [34] combined the BGN password system with differential privacy to achieve fault-tolerant privacy protection and

multifunctional health data aggregation mechanism in cloud assisted WBAN. Shang et al. [35] combined differential privacy with HE and batch verification of signatures, significantly improving overall efficiency. Hu et al. [18] proposed a privacy preserving data aggregation scheme PATM based on smart contracts, which combines secret sharing and differential privacy mechanisms. These schemes assume that all user data is involved. If users are only willing to share a portion of their data, the above multidimensional data aggregation scheme is not applicable.

VIII. CONCLUSION

We proposed a secure and verifiable data aggregation privacy protection scheme in WBANs, called SVODA. This scheme integrated differential privacy to enhance protection without extra computation burden. We further implemented a verification mechanism for tampered data and nonresponsive terminals. Finally, we presented a comprehensive security analysis and performance evaluation. The analysis results show that our scheme is secure, and functional and performance comparisons are significantly superior to existing schemes.

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